

Value Stream Mapping (VSM)

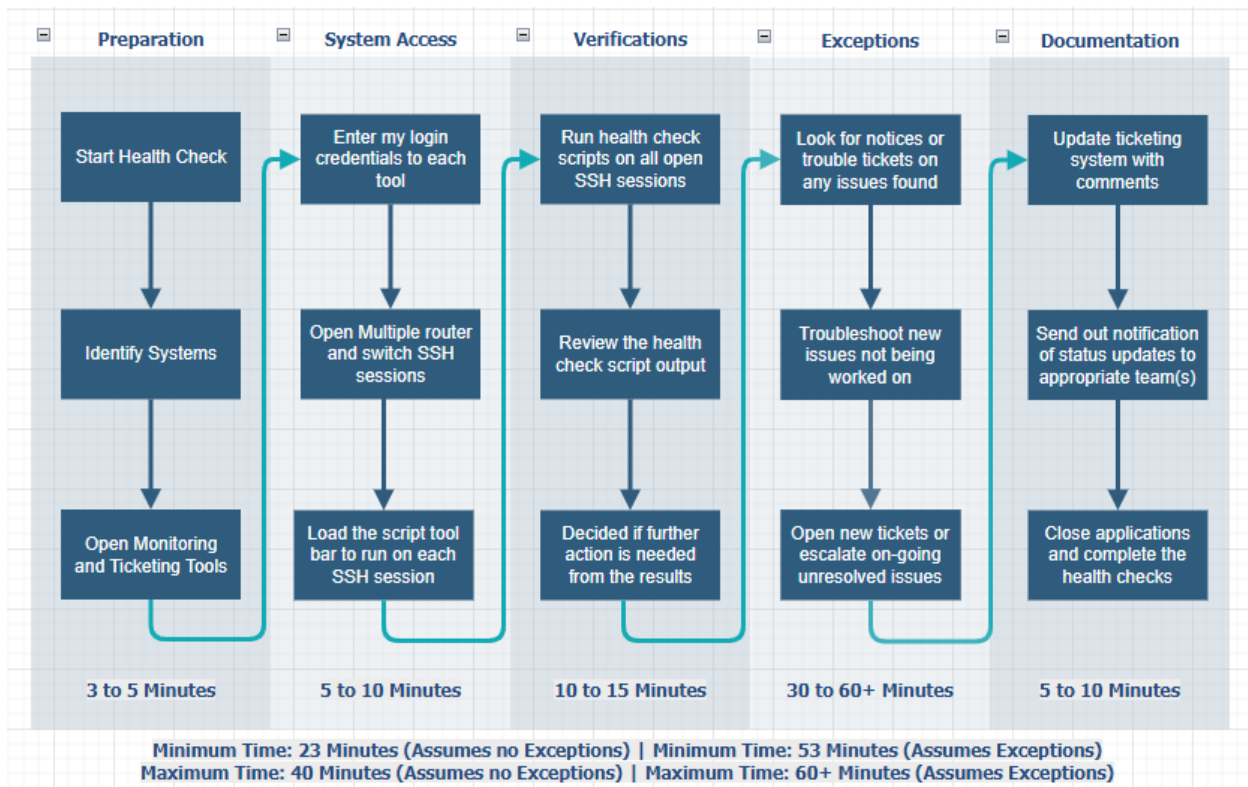
Optimizing Daily Network Health Checks

I want to provide a brief disclaimer based on my professional experience. Having worked in network operations for over twenty years, I have seen firsthand how process improvements and standardization can significantly improve operational efficiency. Finding ways to eliminate unnecessary steps, reduce waiting time, and streamline repetitive tasks is a valuable practice and aligns well with the principles of Lean and Value Stream Mapping.

I have also learned that process improvements should be viewed as opportunities to increase efficiency rather than rigid expectations that every task will always take the same amount of time. In network operations, factors such as application performance, system updates, policy refreshes, hardware issues, or active network outages can all affect how long a health check takes to complete. While an optimized process may establish a realistic baseline under normal operating conditions, real-world situations still require engineers to investigate and respond appropriately.

I chose my daily network health check process because it is something I perform regularly. Network health checks are often performed sequentially by opening each application, waiting for it to load, reviewing the required information, and then moving to the next system. Depending on the engineer and whether any issues are discovered, this process can typically take an hour to complete under normal operating conditions.

I have become more efficient in this workflow by logging into multiple systems in parallel, opening several router sessions at once, and using prebuilt command scripts to gather information concurrently. Provided there are no workstation issues, application delays, policy updates, or actual network problems requiring investigation, the same health checks can typically be completed in approximately 20 to 30 minutes. I feel this is an appropriate example for applying VSM because the improvement comes from reducing idle time rather than removing work that still provides operational value.



Metric	Not Efficient	Efficient
Average Completion Time	60 minutes	20–30 minutes
Waiting on Applications	High	Low
Router Sessions Opened	One at a Time	Simultaneously
Command Execution	Manual Sequential	Scripted Concurrent
Engineer Idle Time	High	Minimal
Process Efficiency	Moderate	High

This optimization removes idle time by opening applications simultaneously, reducing login delays, executing scripted commands across multiple router sessions, and reviewing results in a consolidated workflow. The engineer still performs the same validation activities, but idle time between tasks is reduced. Under normal operating conditions, these improvements reduce the average health checks by 50%.